

The Scenario

Client

My client is a 16-year-old student who has obtained a provisional license and wishes to drive a car when she turns 17. To do so she wishes to obtain her full license which includes a practical test and theoretical test (Refer to Appendix A question 1), however she lacks sufficient resources to effectively study theories.

Problem Statement:

- **Lacks engaging resources to study theory (1.B)**

My client has found existing study materials, such as written notes or videos, both of which prove to be lackluster, boring, and ineffective. The theoretical concept, which include road laws, are overwhelming, and she struggles to retain such information from passive reading, repetition and prosaic content.

- **Lacks enthusiasm to read pages upon pages of notes**

My client is discouraged by the amount of reading required, not only that, but my client finds difficulty in retaining information without engagement or interaction. Traditional methods of studying such as reading pages upon pages fail to engage her and make learning more of a challenge.

- **Lack of Confidence**

Most of the video's my client relies upon for study focus on practical driving skills rather than road theory. This imbalance leaves her unprepared and not confident for the theoretical aspect of the exam thus creating both a lack of confidence and frustration.

Rationale to the solution

The solution is a interactive driving theory and simulation application using Unity and C#. The simulation itself will use Unity's Rigidbody physics engine and collision detection to model realistic vehicle behavior such as steering, braking and acceleration. Driving errors will provide meaningful feedback through event driving scripts such as OnCollisionEnter and triggers which provides visual feedback. Additionally a structured questionbank of road signs and driving laws will be stored to track client's progress. Where the application will use adaptive difficulty algorithms where consecutive correct answers increase difficulty and otherwise reduce it including providing feedback for questions that are incorrect. The program will also include a login system to save and retrieve user progress securely.

Alternative solutions that are viable include:

- A web-based application using HTML, CSS and JavaScript which could deliver quizzes, learning modules and progress tracking through local storage or a server database. However this approach would not support a real-time 3D driving simulator or physics-based interaction
- A native mobile application developed in Java or Swift (Android or IOS) would allow portable learning and testing of theoretical knowledge however the processing power for 3D physics and environment simulation would be limited to individuals that have the processing power.
- A game developed in Unreal Engine using C++ or Blueprints, which would provide advanced physics and visuals however would increase project complexity and is less feasible within the IA time frame

Word Count: 420

Success Criteria (refer to appendix A)

play

- I. The application should include a way for the user to test out their knowledge practically. E.g. simulation
- II. The program should offer environment(s) that simulate real world dynamics
- III. The program should include realistic physics and elements (i.e. lights skid marks etc.)

Test

- IV. The program should include an area for users to test their knowledge
- V. The users should be given the option of topics they want to cover in said tests
- VI. The program should include progressive difficulty that adjusts difficulty

learn

- VII. Users should be able to have a choice in customizing their experience and learning. E.g. different topic focuses
- VIII. The application offers separated modules for road signs, driving laws and rules

login/sign up

- IX. The application includes a working sign in/up page